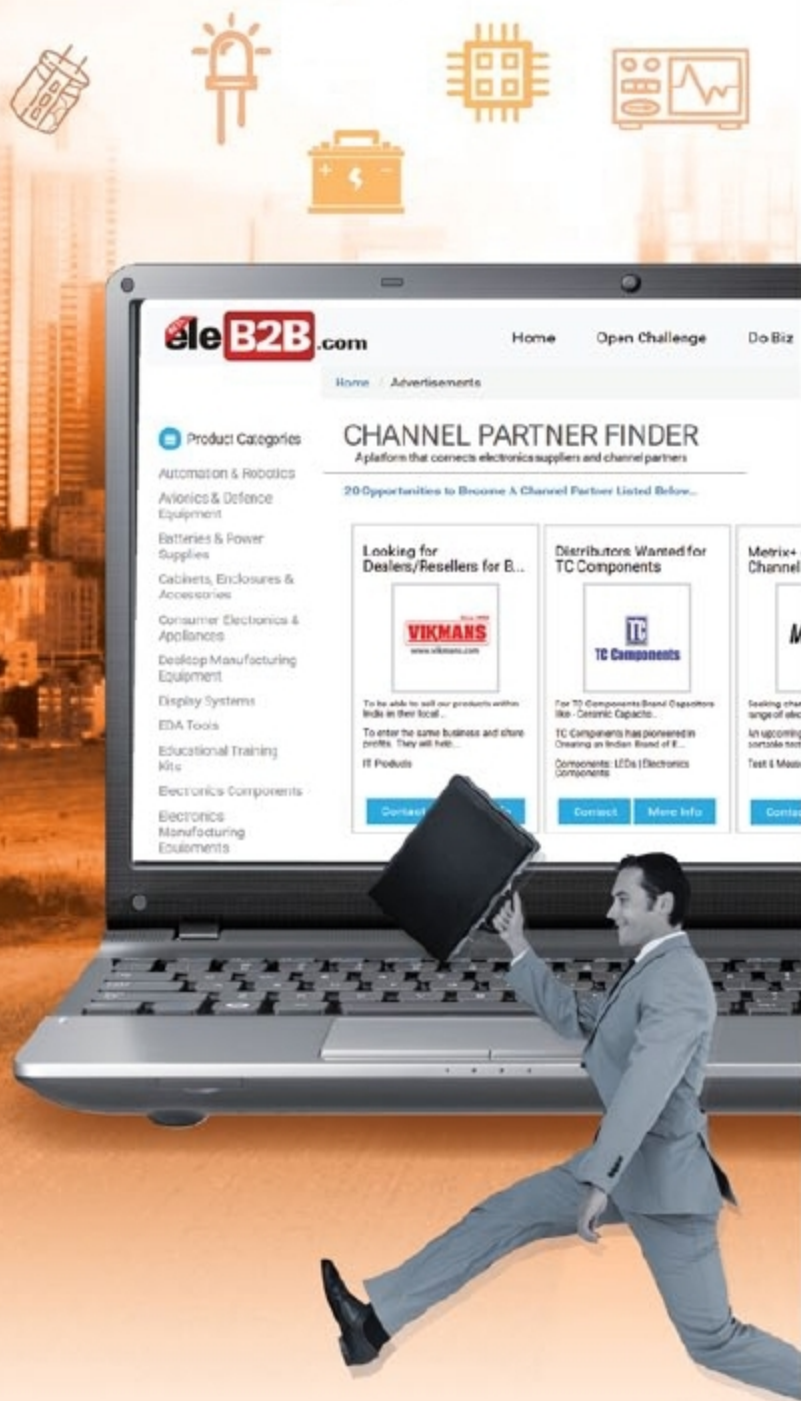


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Channel Partner Finder (CPF) is an online facility introduced by Electronics For You team on its B2B portal: **eleB2B.com**. CPF will enable suppliers of electronics products (materials, components, equipment, etc) to connect with like-minded and interested entrepreneurs and organisations seeking such business opportunities. While the suppliers may be from India or across the globe, the channel partners being engaged are from India.

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and test strategy in place along with calibration. To top it all, improper release mechanism of software and hardware versions adds to the complexity of existing problems.

Fixing the issues

I summarise here a set of issues that need to be addressed if a product has to succeed in the market in its initial release.

1. Hire a consultant for a specific task. Most of the so-called manufacturing consultants have no exposure to engineering and design, and will only help with manufacturing, which is of no use.

2. Do a design analysis and ensure that corner cases are addressed by your design. All components are not identical, and component manufacturers specify the tolerance limits. Your design should be able to work on the extremes of these limits.

3. Due to an increasing number of Internet of Things (IoT) startups and SMEs, most of the hardware measures certain parameters. However, I have not seen a single design that builds calibration against a standard for measurement, and calibration through software rather than using hardware-based calibration.

Calibration of a product is also an involved process and needs proper procedure. Recently, I was looking at a product that measures AC mains current, voltage and ambient temperature. The process automation done by the software was based on these readings. However, when the actual design was reviewed, it had no way to calibrate, and components' tolerance was adding to error, leading to overall errors. In fact, the redesign took three more months and a lot of effort to rectify.

4. Do not get dictated by the manufacturing/assembly house. You must lay out what you want, and they should do what you want. Manufacturing process can also impact your design in terms of bad component handling during placement, improper reflow profile and so on, and you need to understand what happens in the manufacturing line to be in control.

5. The most worrying thing is the use of radio frequency (RF) for communication. I have seen all kinds of RF protocols and frequencies like Wi-Fi, LoRa, LoWPAN, Bluetooth, etc being used. Most designers believe that the modules or devices they use will work as per the specification sheet. The first thing in hardware is that RF has its own mind, and a fairly good understanding of RF is needed if you want your requirements to be met.

Second, RF signals are dependent on the environment they work in. If you install the product with lots of greenery around, chances are you will see below-par performance. Every RF circuit needs a matched antenna, and using a universal antenna will not help. (I have not seen a Bluetooth device that works up to a range of ten metres even today. So, if you are using RF in your design, keep this in mind—no matter how low the emission power.)

Looking at these, you will be surprised to know that a simple EMS will not be helpful, as most of the current vendors do not support engineering. It is up to you as a designer to ensure that engineering is done so your product is successful. **EFY**